The empirical recurrence for OEIS A231399: recovering a conjecture that is not written down

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Abstract

OEIS A231399 records "Empirical recurrence of order 83", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 1594$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 83, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 83 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A231399 is "Number of (3+1)X(n+1) 0..2 arrays with no element unequal to a strict majority of its horizontal, diagonal and antidiagonal neighbors, with values 0..2 introduced in row major order.". It has offset 1 and begins

$$7, 15, 100, 311, 1706, 7844, 35696, 184692, \ \dots$$

The entry states:

> Empirical recurrence of order 83 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 06:27:45, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1594$ vertices, $a(n) = \iota^{\top} M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 1594.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 3188$ exact terms of the model returns order 83 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{83} c_i\, a(n-i),$$

c_{1}	7	c_{22}	−3723533985	c_{43}	5509398872555	c_{64}	−826320744096
c_{2}	32	c_{23}	−25470600967	c_{44}	−714471871549	c_{65}	−42739969792
c_{3}	−249	c_{24}	19661517246	c_{45}	−5362969936535	c_{66}	392670443712
c_{4}	−740	c_{25}	56660225263	c_{46}	2941440627589	c_{67}	25497507072
c_{5}	5032	c_{26}	−72340888817	c_{47}	4429376138506	c_{68}	−164966043520
c_{6}	11647	c_{27}	−101165100137	c_{48}	−4892967786613	c_{69}	−9729676032
c_{7}	−68903	c_{28}	209930923365	c_{49}	−3110685787799	c_{70}	60112793856
c_{8}	−126877	c_{29}	130637873897	c_{50}	6156597245440	c_{71}	1728013312
c_{9}	691230	c_{30}	−501388564584	c_{51}	1854055231779	c_{72}	−18453561344
c_{10}	1012117	c_{31}	−66074702759	c_{52}	−6566389988226	c_{73}	435073024
c_{11}	−5400451	c_{32}	1003314260142	c_{53}	−924507878338	c_{74}	4596908032
c_{12}	−6104272	c_{33}	−230100462355	c_{54}	6172767974340	c_{75}	−421498880
c_{13}	34206596	c_{34}	−1691290597048	c_{55}	368377793728	c_{76}	−886513664
c_{14}	27966599	c_{35}	911431209389	c_{56}	−5178763130224	c_{77}	149745664
c_{15}	−179987948	c_{36}	2388313069019	c_{57}	−107382073560	c_{78}	123871232
c_{16}	−93357851	c_{37}	−2039633277254	c_{58}	3882909115384	c_{79}	−30916608
c_{17}	797569355	c_{38}	−2760931356302	c_{59}	29841017104	c_{80}	−11173888
c_{18}	181018294	c_{39}	3460786074640	c_{60}	−2597899031200	c_{81}	3670016
c_{19}	−2993334443	c_{40}	2439949338534	c_{61}	−33508067728	c_{82}	491520
c_{20}	208469739	c_{41}	−4778324427842	c_{62}	1550554324064	c_{83}	−196608
c_{21}	9513564459	c_{42}	−1232457043742	c_{63}	47198919200		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 83, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $83+4$ terms removed returns order 83 as well, so $\deg q_{\min} = 83 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 22 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $83+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 83 that this sequence satisfies — and that family turns out to have exactly one member.

References

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